Video Quality Prediction

Graph Neural Networks for Advertising Popularity

GROUP 6 / TOPIC 3

01 RESEARCH OBJECTIVE

Test whether relationships among advertising creative characteristics improve video-popularity prediction over feature-based models. Identify combinations of message, hook, pacing and creative mechanic associated with stronger public response, then translate supported patterns into evidence-linked creative suggestions.

02 DATA AND FEASIBILITY

The supplied archive contains 219 unique videos across 14 current brand packages after retaining the latest HSBC and Hang Seng versions. Only 20 McDonald's cases currently contain views/likes. We propose to reconcile schemas, remove duplicate records and identify related campaign variants, audit extracted evidence, and obtain response labels at a common documented snapshot through approved sources. Primary target: log(1 + views); fit_score is not a popularity label. With insufficient labels, supervised work remains a pilot and the main analysis will be exploratory.

03 PROPOSED METHOD

Use each video as a node, with encoded creative categories, text embeddings and available metadata as features. Test a lightweight GraphSAGE model on feature-similarity edges against a mean predictor, ridge regression and a small MLP using matched inputs. Review candidate edges as heuristics and test their contribution. Inspect/reproduce the referenced VideoGNN baseline if its code and data are supplied; they are absent from the current archive.

04 EVALUATION AND VALIDITY

Reserve 20% of labelled data for testing, grouping related campaign variants and duplicates; tune on folds within the remaining 80%. Fit preprocessing and neighbour rules on training data only, remove direct/derived response information (including text), and attach held-out videos using feature-only links to training references. Report MAE/RMSE on log(1 + views) and Spearman rank correlation, with sample sizes and uncertainty. Compare metadata-only versus creative features and graph versus no-graph models. Check exposure age, duration, platform and brand/channel context where available.

05 WORK PLAN AND DELIVERABLES

Stage 1: reconcile the inventory, evidence and label coverage. Stage 2: build baselines and a small GNN if labels permit. Stage 3: evaluate held-out predictions, ablations and failure cases. Stage 4: prepare a reproducible report, an inspectable results/graph website, and one creative case study linking suggested changes to source examples. Dates and individual responsibilities will be agreed within Group 6 against the course deadlines.

06 SCOPE AND RESPONSIBLE INTERPRETATION

Findings describe associations in a small observational sample; views do not establish sales, brand recognition or causal creative effects. Preserve source provenance, flag weak evidence, and respect brand-owned material when proposing changes. No model result or improvement is claimed at the planning stage.